PHISHING ANALYSIS REPORT

Target - https://github.com/rf-peixoto/phishing_pot/blob/main/email/sample-1001.eml

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Phishing Report

Executive Summary

Phishing is a cybercrime where attackers masquerade as reputable sources via email, text, or phone to steal sensitive information like passwords, credit card details, and personal data. These fraudulent messages often create a false sense of urgency, urging victims to click malicious links or download attachments, resulting in financial loss or identity theft.

Background

How Phishing Works

Phishing attacks typically involve fraudulent communication, often through emails, messages, or websites that appear legitimate. The attacker tricks the victim into clicking a malicious link, downloading an attachment, or providing confidential information. Once the victim responds, the attacker gains access to sensitive data or systems.

Types of Phishing

1. **Email Phishing:** Mass emails sent to many users posing as legitimate organizations.
2. **Spear Phishing:** Targeted attacks aimed at specific individuals or organizations.
3. **Smishing:** Phishing conducted via SMS or text messages.
4. **Vishing:** Voice-based phishing using phone calls.
5. **Clone Phishing:** A legitimate message is copied and altered with malicious links or attachments.

Recent trends in phishing

AI phishing: AI phishing uses generative artificial intelligence (AI) tools to create phishing messages. These tools can generate tailored emails and text messages that lack spelling errors, grammatical inconsistencies and other common red flags of phishing attempts.

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Quishing: uses fake QR codes embedded in emails and text messages or posted in the real world. Quishing allows hackers to hide malicious websites and software in plain sight.

Hybrid vishing: These attacks combine voice phishing with other methods to evade spam filters and gain victims' trust.

Common Techniques Used

- Creating fake websites that resemble official ones
- Using urgent or alarming messages to pressure victims
- Spoofing email addresses or phone numbers
- Embedding malicious links or attachments

Impacts of Phishing

- Financial loss
- Identity theft
- Unauthorized access to accounts
- Data breaches in organizations
- Damage to reputation and trust

Prevention Measures

- Verify the sender before responding to messages
- Avoid clicking on suspicious links or attachments
- Use strong, unique passwords and enable two-factor authentication
- Keep software and antivirus updated
- Educate users about cybersecurity awareness

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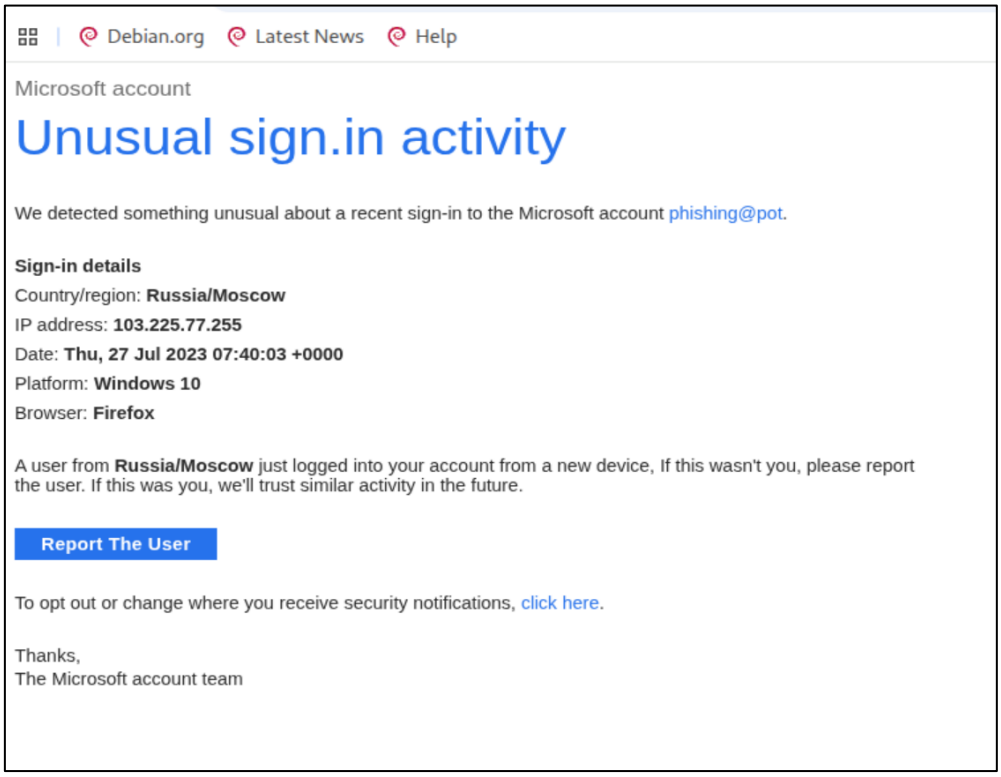
Phishing Analysis

Sample Email for analysis

Target: https://github.com/rf-peixoto/phishing_pot/blob/main/email/sample-1001.eml



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Microsoft account

Unusual sign.in activity

We detected something unusual about a recent sign-in to the Microsoft account phishing@pot.

Sign-in details
Country/region: **Russia/Moscow**
IP address: **103.225.77.255**
Date: **Thu, 27 Jul 2023 07:40:03 +0000**
Platform: **Windows 10**
Browser: **Firefox**

A user from **Russia/Moscow** just logged into your account from a new device, If this wasn't you, please report the user. If this was you, we'll trust similar activity in the future.

Report The User

To opt out or change where you receive security notifications, click here.

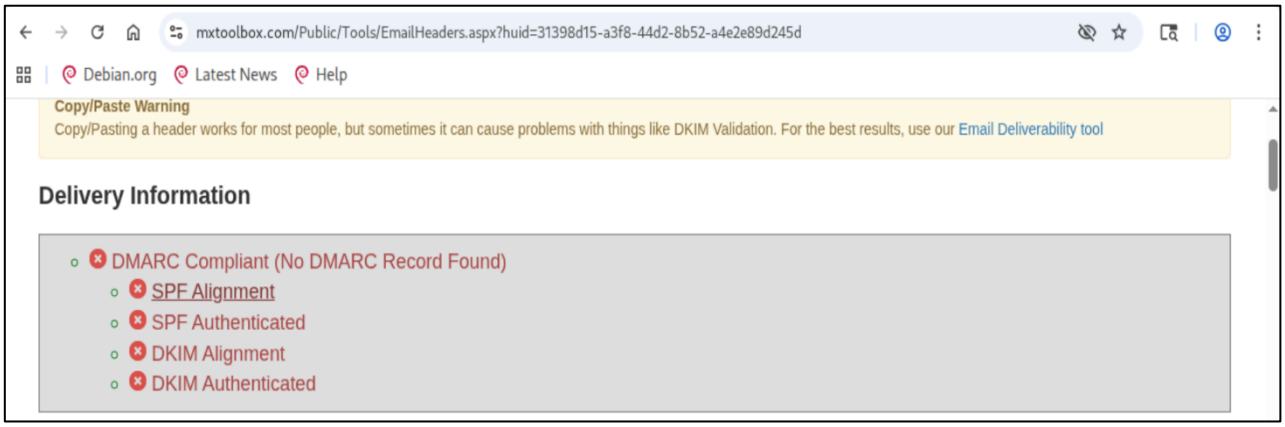
Thanks,
The Microsoft account team

Figure 1: Sample email 1

Analysis Methodology

Email header analysis

Below are the screenshots of result of email header analysis using MXToolbox



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Authentication Results:

The above evidence shows that

1. spf=none, dkim=none, dmarc=permerror, SPF=none
   This means that there's no permission for that server to send email.
   SPF normally lists allowed servers and DMARC uses SPF + DKIM to verify sender. Here, the authentication is broken.
   Conclusion- Strong indicator of spoofing
2. DKIM = none
   This means no digital signature.
   Whereas, legit companies always sign emails
   Conclusion- Message integrity NOT verified
3. There are inconsistencies like:
   "From: access-accsecurity.com", "Return-Path: nonkfrgr.co.uk".
   These should match in legitimate emails. But there's a mismatch.
   Conclusion- spoofing attempt

Headers Found

Header Name	Header Value
Authentication-Results	spf=none (sender IP is 89.144.9.87) smtp.mailfrom=nonkfrgr.co.uk; dkim=none (message not signed) header.d=none;dmarc=permerror action=none header.from=access-accsecurity.com;
Received-SPF	None (protection.outlook.com: nonkfrgr.co.uk does not designate permitted sender hosts)
X-IncomingTopHeaderMarker	OriginalChecksum:581851118E8AB60FB557B9652148A1D2ACF0E25191DB3370DD6E91B6F7AF2E34;UpperCasedChecksum:650D4BED98031B7AE86C487AC7AF30C830E0CD64B4F923A90386050DC65A28C4;SizeAsReceived:396;Count:12
From	Microsoft account team ._.<no-reply@access-accsecurity.com>
Subject	Microsoft account unusual signin activity
To	phishing@pot
Content-Length	18708448
Date	Thu, 27 Jul 2023 07:40:03 +0000
Reply-To	solutionteamrecognizd02@gmail.com
Content-Type	text/html; charset="UTF-8"
Content-Transfer-Encoding	8bit
X-IncomingHeaderCount	12
Message-ID	<412928a2-0aad-4b27-959a-5df404e1f07d@VI1EUR06FT066.eop-eur06.prod.protection.outlook.com>
Return-Path	bounce@nonkfrgr.co.uk
X-MS-Exchange-Organization-ExpirationStartTime	27 Jul 2023 07:40:03.4716 (UTC)
X-MS-Exchange-Organization-ExpirationStartTimeReason	OriginalSubmit
X-MS-Exchange-Organization-ExpirationInterval	1:00:00:00.0000000

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X-MS-Exchange-Organization-AuthSource	VI1EUR06FT066.eop-eur06.prod.protection.outlook.com
X-MS-Exchange-Organization-AuthAs	Anonymous
X-MS-UserLastLogonTime	7/27/2023 7:32:58 AM
X-MS-Office365-Filtering-Correlation-Id	92ccd2d7-4aa5-4320-c208-08db8e74b0af
X-MS-Exchange-EOPDirect	true
X-Sender-IP	89.144.9.87
X-SID-PRA	NO-REPLY@ACCESS-ACCSECURITY.COM
X-SID-Result	NONE
X-MS-Exchange-Organization-PCL	2
X-MS-Exchange-Organization-SCL	5
X-Microsoft-Antispam	BCL:6;
X-MS-Exchange-CrossTenant-OriginalArrivalTime	27 Jul 2023 07:40:03.3778 (UTC)
X-MS-Exchange-CrossTenant-Network-Message-Id	92ccd2d7-4aa5-4320-c208-08db8e74b0af
X-MS-Exchange-CrossTenant-Id	84df9e7f-e9f6-40af-b435-aaaaaaaaaaaa
X-MS-Exchange-CrossTenant-AuthSource	VI1EUR06FT066.eop-eur06.prod.protection.outlook.com
X-MS-Exchange-CrossTenant-AuthAs	Anonymous

4. The" X-MS-Exchange-Organization-SCL: 5" indicates Spam confidence level is high; Microsoft already suspects it.

Headers Found

Header Name	Header Value
Authentication-Results	spf=none (sender IP is 89.144.9.87) smtp.mailfrom=nonkfrgr.co.uk; dkim=none (message not signed) header.d=none;dmarc=permerror action=none header.from=access-accsecurity.com;
Received-SPF	None (protection.outlook.com: nonkfrgr.co.uk does not designate permitted sender hosts)
X-IncomingTopHeaderMarker	OriginalChecksum:581851118E8AB60FB557B9652148A1D2ACF0E25191DB3370DD6E91B6F7AF2E34;UpperCasedChecksum:650D4BED98031B7AE86C487AC7AF30C830E0CD64B4F923A90386050DC65A28C4;SizeAsReceived:396;Count:12
From	Microsoft account team ,_<no-reply@access-accsecurity.com>
Subject	Microsoft account unusual signin activity
To	phishing@pot
Content-Length	18708448
Date	Thu, 27 Jul 2023 07:40:03 +0000
Reply-To	solutionteamrecognizd02@gmail.com
Content-Type	text/html; charset="UTF-8"
Content-Transfer-Encoding	8bit
X-IncomingHeaderCount	12
Message-ID	<412928a2-0aad-4b27-959a-5df404e1f07d@VI1EUR06FT066.eop-eur06.prod.protection.outlook.com>
Return-Path	bounce@nonkfrgr.co.uk
X-MS-Exchange-Organization-ExpirationStartTime	27 Jul 2023 07:40:03.4716 (UTC)
X-MS-Exchange-Organization-ExpirationStartTimeReason	OriginalSubmit
X-MS-Exchange-Organization-ExpirationInterval	1:00:00:00.0000000

5. "Reply-To: Gmail address": It claims Microsoft but replies go to Gmail attacker

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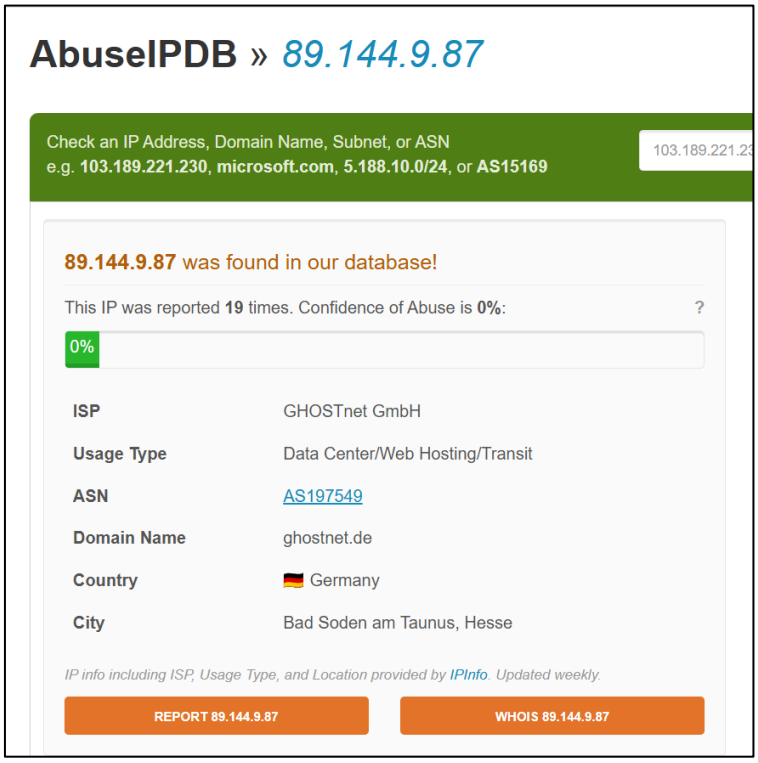
IP intelligence test

- IP address verification using AbuseIPDB



The email claims sign-in from Russia while the IP address seems to be from Germany. The country mismatch is an evidence that the email might be scam.

Infrastructure Analysis

The domains found - access-accsecurity.com seem to be suspicious, nonkfrgr.co.uk seems to be unrelated.

Social Engineering Indicators

Urgency ("unusual sign-in"), Fear (foreign location), Fake branding (Microsoft), Malicious response path

Conclusion

All the above analysis results conclude that the email analysed is a phishing email.